

Skills Summary

Unit Rates with Fractions

Use this reference while completing your worksheet or when studying.

What a unit rate is. A unit rate says how much there is for exactly *one* — one hour, one pound, one batch. Written as a division, it is *amount* $\div$ *number of units*.

Dividing by a fraction: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ — keep the first fraction, flip the second, multiply.

Mixed numbers first. Rewrite any mixed number as an improper fraction *before* you flip anything.

1. Find a unit rate from two fractions

Put the amount on top and the number of units underneath, then multiply by the reciprocal. Dividing by a fraction below one makes the answer *bigger* — that is not a mistake, it is what “per one whole unit” means.

Example: A snail crawls $\frac{3}{4}$ of a metre in $\frac{1}{2}$ of an hour. How many metres per hour?

Step 1. “Per hour” names one hour as the unit, so divide the distance by the time rather than comparing them any other way.

Step 2. Flip the second fraction and multiply: $\frac{3}{4} \times \frac{2}{1} = \frac{3}{2}$.

The unit rate is $\frac{3}{2}$ metres per hour.

Try it: A pump moves $\frac{2}{5}$ of a tank in $\frac{1}{4}$ of an hour. How much of a tank per hour?

check: $\frac{5}{8}$

2. Compare two rates by their unit rates

Two rates with different amounts *and* different times cannot be compared as they stand. Turn each into a unit rate — the same “per one” for both — and then the two numbers can be compared directly.

Example: Ali cycles $\frac{3}{8}$ km in $\frac{1}{4}$ h; Bo cycles $\frac{4}{5}$ km in $\frac{2}{5}$ h. Who is faster?

Step 1. Different distances over different times share no common footing, so rewrite both as kilometres per one hour before comparing.

Step 2. Ali: $\frac{3}{8} \times \frac{4}{1} = \frac{3}{2}$. Bo: $\frac{4}{5} \times \frac{5}{2} = 2$.

Ali $\frac{3}{2}$, Bo **2**, so Ali's rate $<$ Bo's rate.

Try it: Kai runs $\frac{2}{3}$ of a mile in $\frac{1}{2}$ of an hour; Mia runs $\frac{3}{4}$ of a mile in $\frac{1}{2}$ of an hour. Find both unit rates and write $>$ or $<$ between them.

check: Kai $\frac{4}{3}$, Mia $\frac{3}{2}$, Kai $>$ Mia

3. Find a unit rate in a word problem

Find the two quantities the story pairs up, and ask which one the question wants *one* of. That one goes underneath. Then divide as usual, and label the answer with both units so you can hear whether it makes sense.

Example: A pump empties $\frac{5}{6}$ of a tank in $\frac{2}{3}$ of an hour. What fraction of a tank per hour?

Step 1. The question asks for one hour's worth, so the hours go underneath and the tank amount on top.

Step 2. $\frac{5}{6} \times \frac{3}{2} = \frac{5}{4}$.

That is $\frac{5}{4}$ of a tank per hour.

Try it: A printer finishes $\frac{3}{5}$ of a job in $\frac{1}{4}$ of an hour. How many jobs per hour?

check: $\frac{5}{12}$

(!) **Watch out:** the costliest slip is dividing the wrong way round. Before you flip, say the answer out loud with its units — “miles per hour” puts miles on top and hours underneath. “Hours per mile” is a real quantity too, just not the one the question asked for.